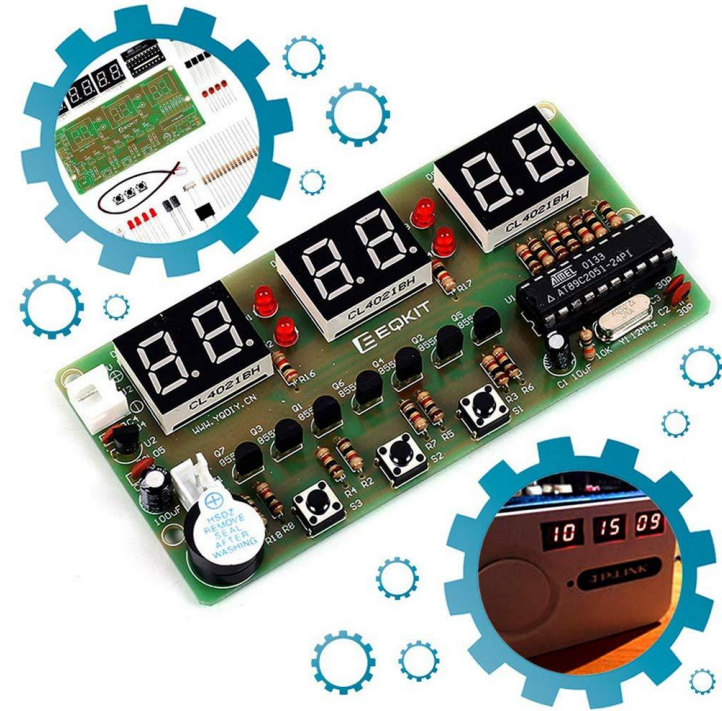




Dr. Hiran Ekanayake

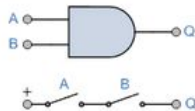
DIGITAL ELECTRONIC FUNDAMENTALS – PART 1

References

- Tutorials on Logic Gates
 - <https://www.electronics-tutorials.ws/category/logic>

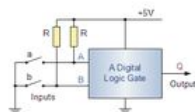


Logic Gates (12)



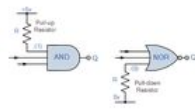
Universal Logic Gates

Individual logic gates can be connected together to form a variety of different switching functions and combinational logic circuits. As we have seen through this Digital Logic tutorial section, the three most basic logic gates are the: AND, OR and NOT gates, and given this set ...



Pull-up Resistors

Digital logic gates can be used for connection to external circuits or devices but care must be taken to ensure that their inputs or outputs function correctly and provide the expected switching condition. Modern digital logic gates, IC's and micro-controllers contain many input...



Digital Logic Gates Summary

We have also seen that each gate has an opposite or complementary form of itself in the form of the NAND Gate, the NOR Gate and the Buffer respectively, and that any of these individual gates can be connected together to form more complex Combinational Logic circuits. We have al...

Lesson Outline

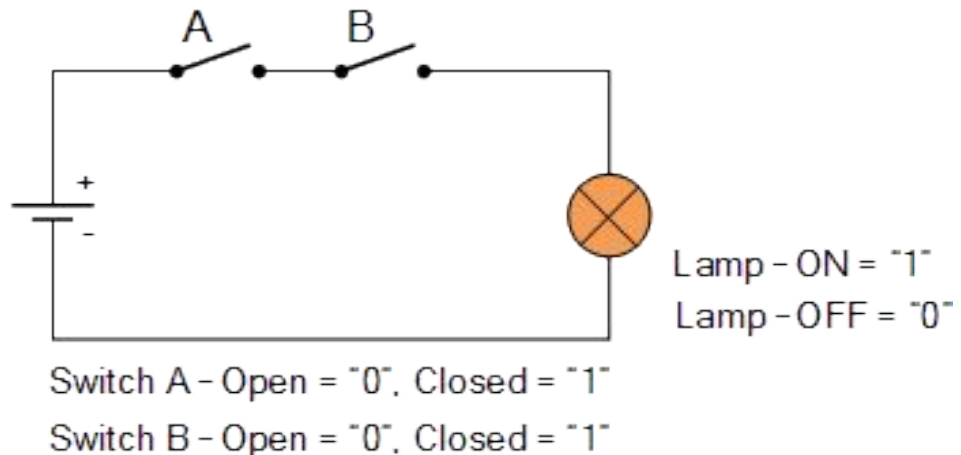
- AND, OR, and NOT logic functions
- Digital logic ICs

LOGIC AND

Logic AND Function

- Describe the logic AND function.
 - The Logic AND Function output is only true when all of its inputs are true, otherwise the output is false.
 - It states that two or more events must occur together and at the same time for an output action to occur. The order in which these actions occur is unimportant.
- Give an example to demonstrate the logic AND function.

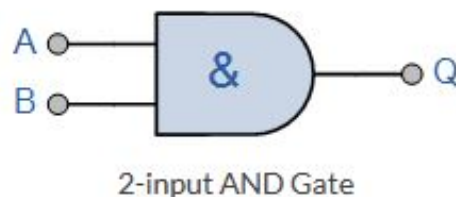
In the given series circuit, both switch A AND switch B must be closed (Logic "1") in order to put the lamp on.



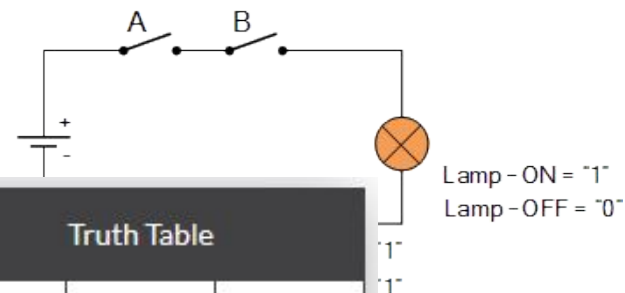
Logic AND Function

- Give the truth table of the logic AND function.

Switch A	Switch B	Output	Description
0	0	0	A and B are both open, lamp OFF
0	1	0	
1	0	0	
1	1	1	
Boolean Expression (A AND B)			



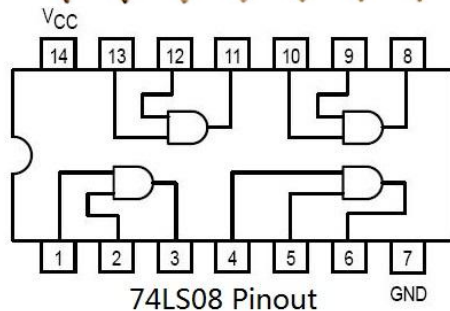
Boolean Expression $Q = A.B$



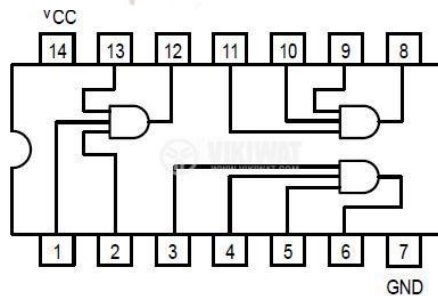
Truth Table		
B	A	Q
0	0	0
0	1	0
1	0	0
1	1	1
Read as A AND B gives Q		

Logic AND Function

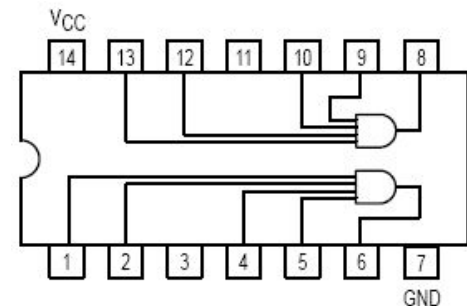
- Which electronic component provides logic AND function?
 - IC (integrated circuit) packages



Quad 2-Input AND Gates



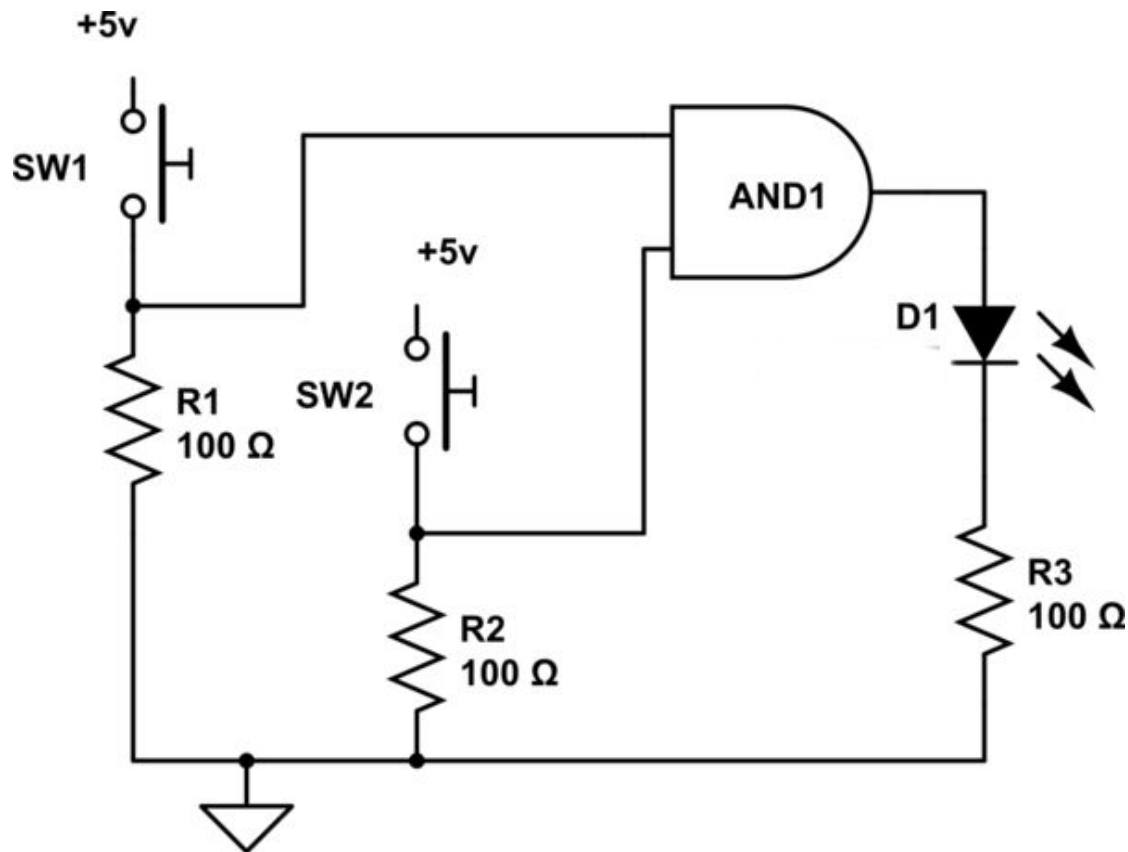
Triple 3-Input AND Gates



Dual 4-Input AND Gates

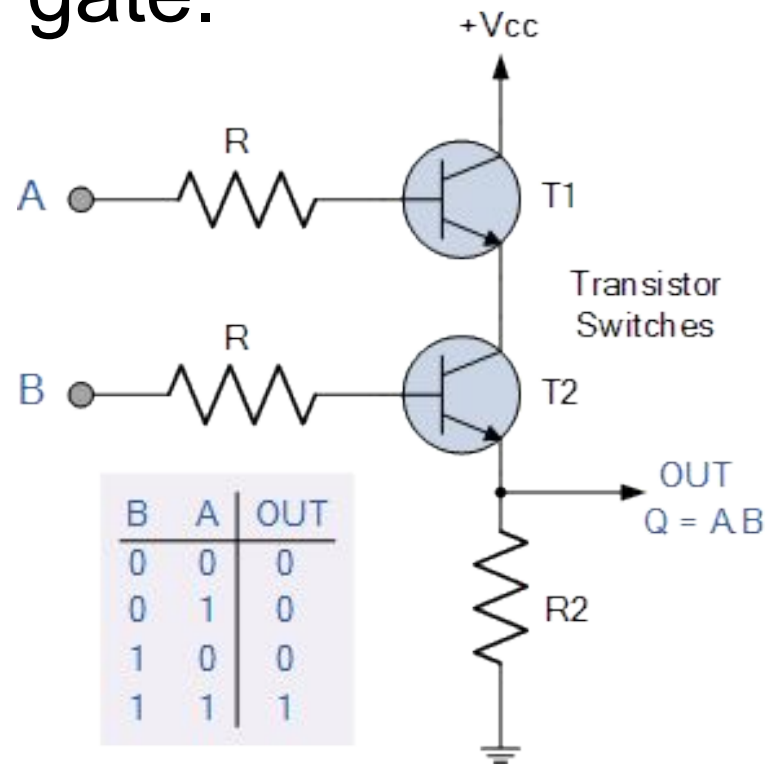
Logic AND Function

- Give a test circuit to test a 2-input AND gate.



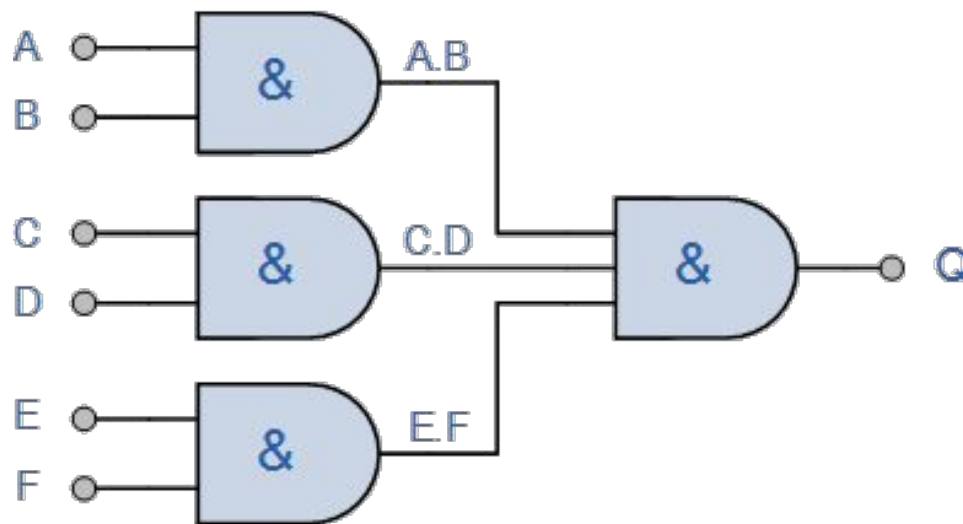
Logic AND Function

- Give an equivalent transistor-based circuit for the AND gate.



Logic AND Function

- How would you construct a 6-input AND gate using 2-input and/or 3-input AND gates?



**Boolean
Expression:** $Q = (A.B).(C.D).(E.F)$

Question 01

A safety system for a machine has 3 conditions that must ALL be met before the machine is allowed to start:

- **A** = Emergency stop button is released
- **B** = Safety guard is in place
- **C** = Operator key is turned

Using AND gates, design a circuit for the given scenario.

Draw the gate arrangement and write the Boolean expression. Draw the truth table.

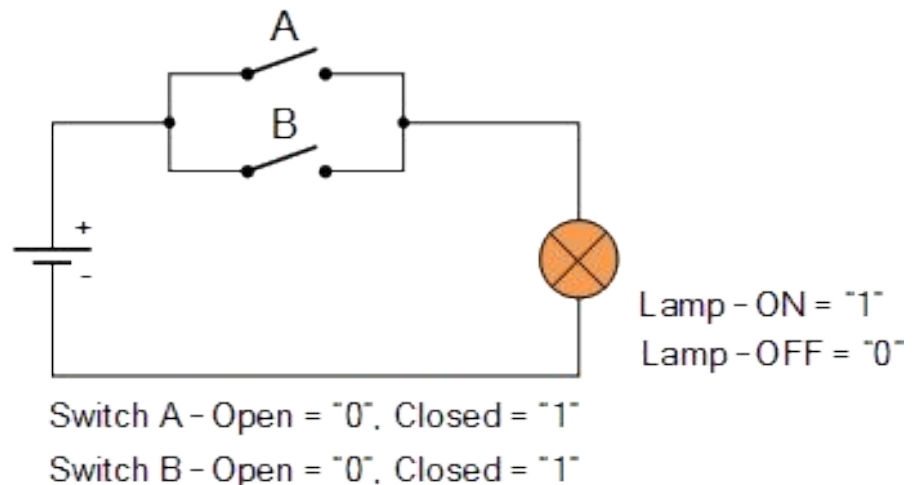
Hint: You only have 2-input AND gates available.

LOGIC OR

Logic OR Function

- Describe the logic OR function.
 - The Logic OR function output is only true if one or more of its inputs are true, otherwise the output is false.
 - Also called “Inclusive-OR”
- Give an example to demonstrate the logic OR function.

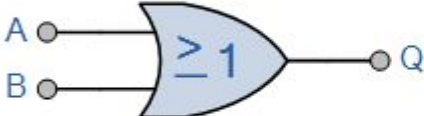
Here the two switches A and B are connected in parallel and either Switch A OR Switch B can be closed in order to put the lamp on.

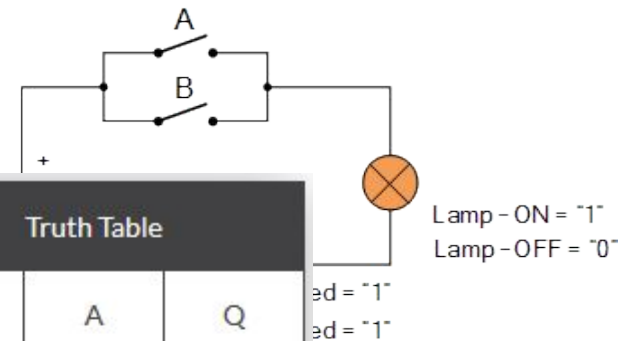


Logic OR Function

- Give the truth table of the logic OR function.

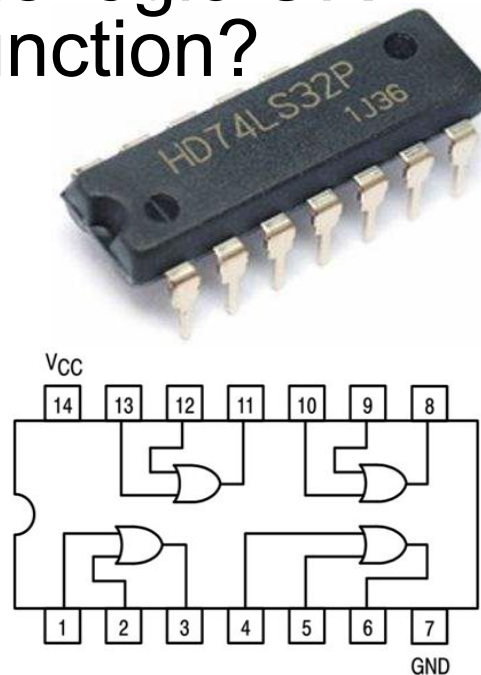
Switch A	Switch B	Output	Description
0	0	0	A and B are both open, lamp OFF
0	1	1	
1	0	1	
1	1	1	
Boolean Expression (A OR B)			

Symbol	Truth Table		
 2-input OR Gate	B	A	Q
	0	0	0
	0	1	1
	1	0	1
	1	1	1
Boolean Expression $Q = A + B$		Read as A OR B gives Q	



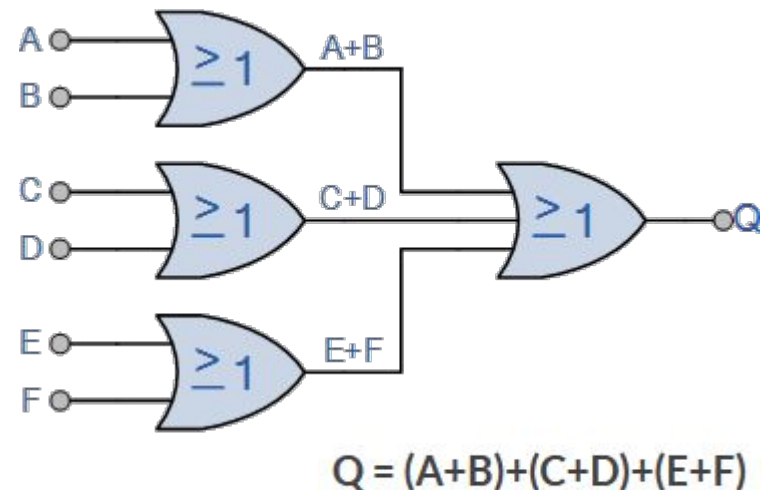
Logic OR Function

- Which IC packages provide the logic OR function?



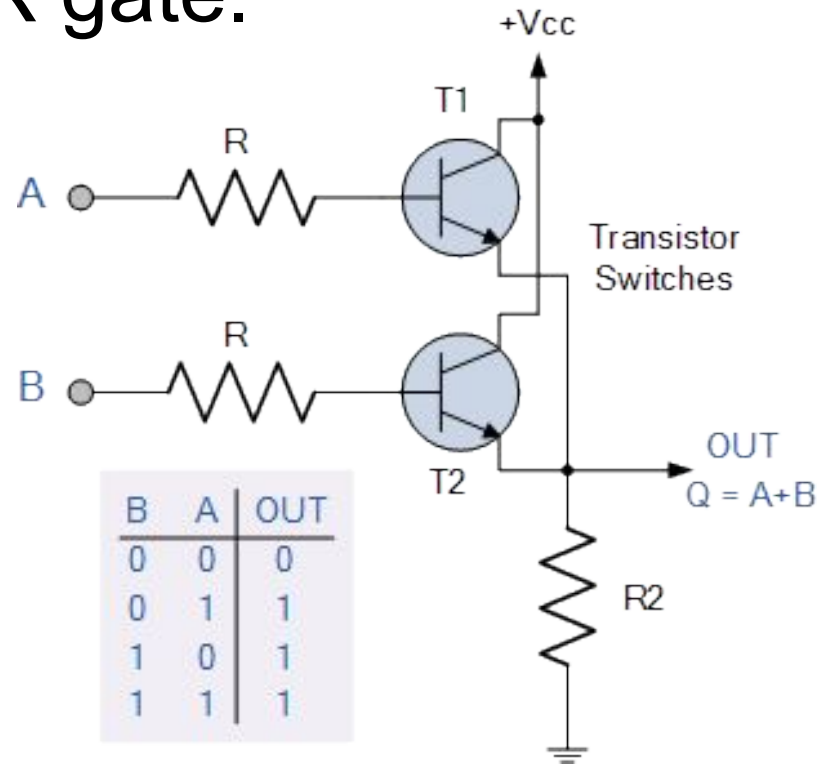
Quad 2-Input OR Gates

- How would you construct a 6-input OR gate using 2-input and/or 3-input OR gates?



Logic OR Function

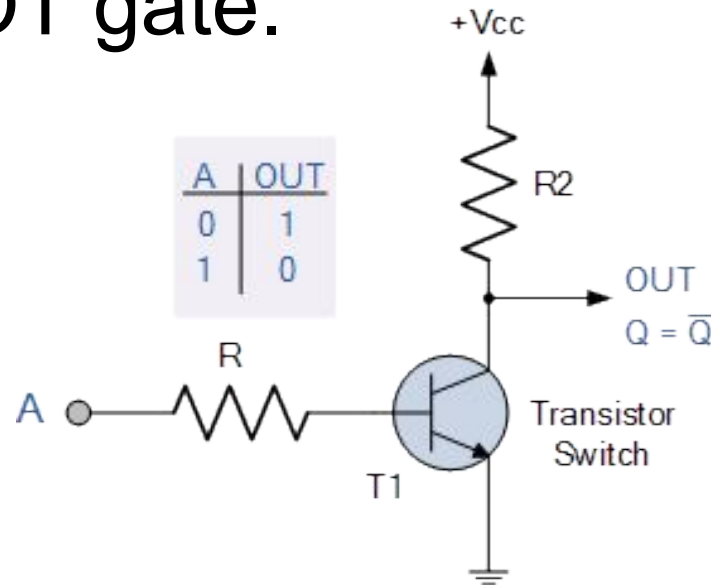
- Give an equivalent transistor-based circuit for the OR gate.



LOGIC NOT (INVERTER)

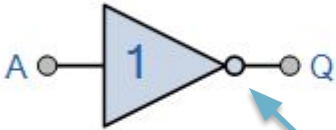
Logic NOT Function

- Describe the logic NOT function.
 - It “inverts” (complements) its input signal.
- Give an equivalent transistor-based circuit for the NOT gate.



Logic NOT Function

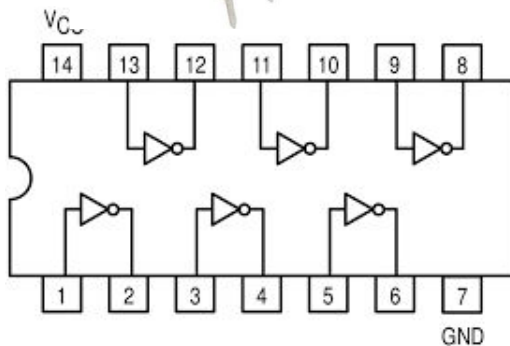
- Give the truth table of the logic NOT function.

Symbol	Truth Table	
 Inverter or NOT Gate	A	Q
	0	1
	1	0
Boolean Expression $Q = \text{not } A \text{ or } \bar{A}$		Read as inverse of A gives Q

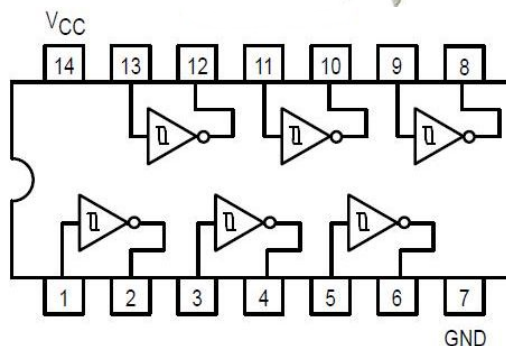
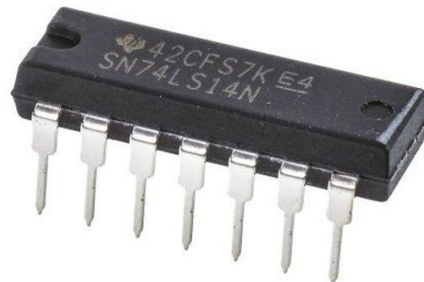
Inversion
Bubble

Logic NOT Function

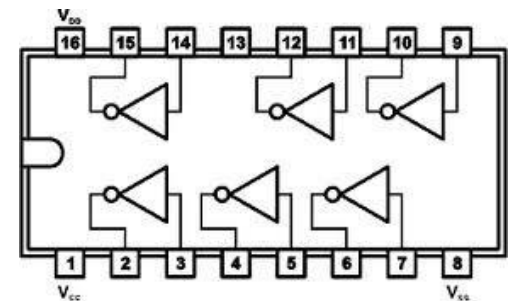
- Which IC packages provide the logic NOT function?



Hex Inverter



Hex Schmitt Inverter



Hex Inverter

Question 02

A circuit is described by:

$$Q = \overline{A \cdot B} + C$$

Using only AND, OR, and NOT gates:

- Draw the logic diagram.
- Construct the truth table.
- Determine for which input combinations $Q=1$

Question 03

A security system should trigger an alarm when:

- The door sensor (**D**) detects the door is open, **OR**
- The window sensor (**W**) detects the window is open,

AND

- The system is armed (**A = 1**).

Using only AND, OR, and NOT gates:

1. Write the Boolean expression for the alarm output Q.
2. Draw the logic circuit and truth table.

Question 04

A vault opens when:

- Key switch (K) is ON

AND

- Either password is correct (P) OR fingerprint is verified (F)

AND

- Alarm state (A) is OFF

AND

- Maintenance mode (M) is OFF

Tasks

1. Derive the Boolean expression.
2. Draw the logic circuit using only AND, OR, and NOT gates.
3. Create the truth table.
4. Determine the output when:
 - K=1, P=0, F=1, A=0, M=0
 - K=1, P=1, F=0, A=1, M=0

Question 05

Write the boolean expression for the given truth table

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

DIGITAL ICS

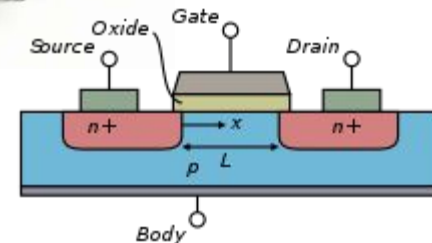
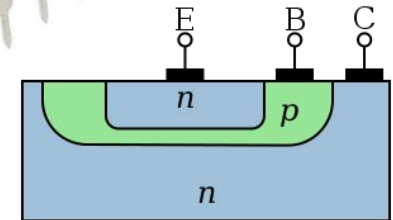
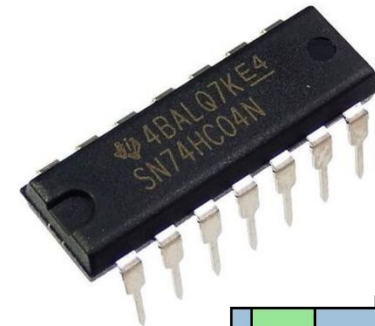
Digital Logic Gates

- What is the difference between 74XX and 4XXX IC packages?
 - Are they compatible with each other?
- What is the convention used in the labelling of IC packages?



Main Families of Logic Gates

- Transistor-Transistor Logic (TTL)
 - Mainly the 7400 series of chips
 - Use NPN and PNP type Bipolar Junction Transistors to implement the logic
- Complementary Metal-Oxide-Silicon (CMOS)
 - Mainly the 4000 series of chips
 - Use complementary MOSFET or JFET type Field Effect Transistors



Classification of ICs

- Small Scale Integration or (SSI)
 - 10 or few transistors in a package to form simple gates like AND, OR, and NOT
- Medium Scale Integration or (MSI)
 - 10 – 100 transistors to perform digital operations such as adders, decoders, counters, flip-flops and multiplexers
- Large Scale Integration or (LSI)
 - 100 – 1,000 transistors to perform specific digital operations such as I/O chips, memory, arithmetic and logic units

Classification of ICs

- Very-Large Scale Integration or (VLSI)
 - 1,000 – 10,000 transistors to perform computational operations such as processors, large memory arrays and programmable logic devices
- Super-Large Scale Integration or (SLSI)
 - 10,000 – 100,000 transistors to perform computational operations such as microprocessor chips, micro-controllers, basic PICs and calculators

Classification of ICs

- Ultra-Large Scale Integration or (ULSI)
 - More than 1 million transistors in a package, used in computers CPUs, GPUs, video processors, micro-controllers, FPGAs and complex PICs

E.g., Intel Core2 Duo
Processor E8500 uses
410 million transistors

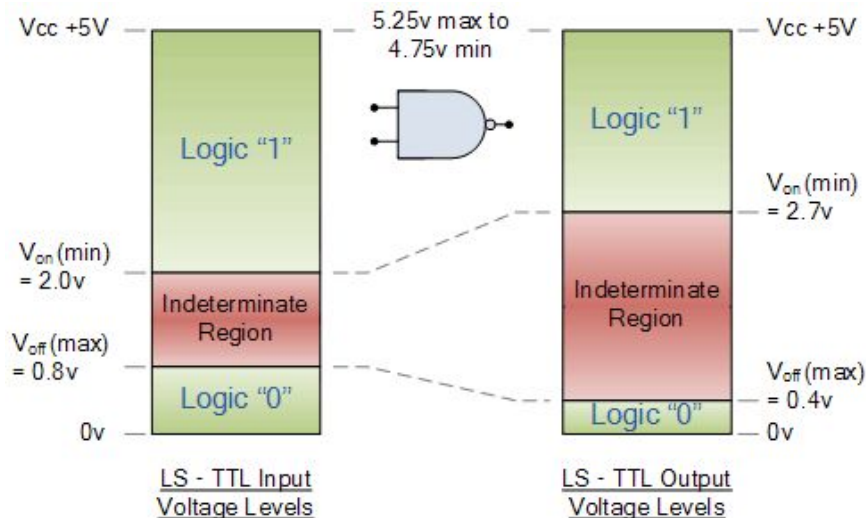
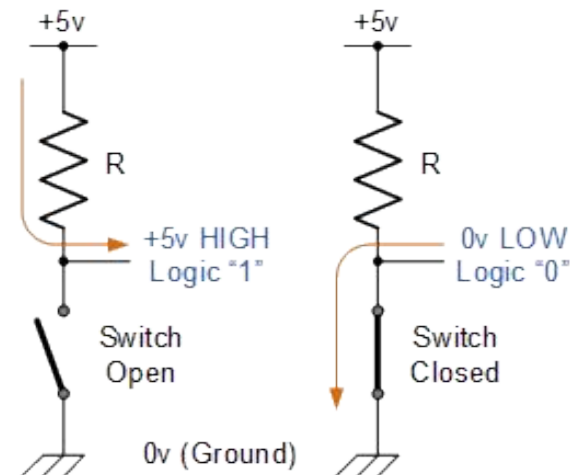
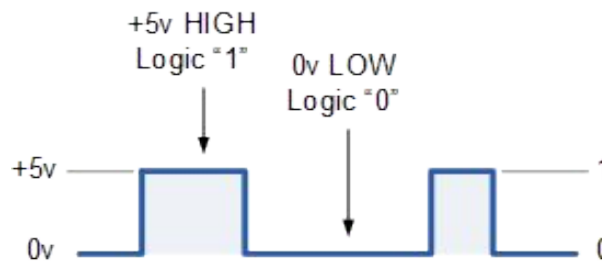


Summary

Level	Name	Transistors	What's inside
SSI	Small Scale	< 10	Basic gates — AND, OR, NOT
MSI	Medium Scale	10–100	Adders, counters, flip-flops, decoders
LSI	Large Scale	100–1,000	Memory chips, I/O controllers
VLSI	Very Large Scale	1,000–10,000	Early processors, large memory
SLSI	Super Large Scale	10,000–100,000	Microcontrollers, basic PICs, calculators
ULSI	Ultra Large Scale	> 1,000,000	Modern CPUs, GPUs, FPGAs

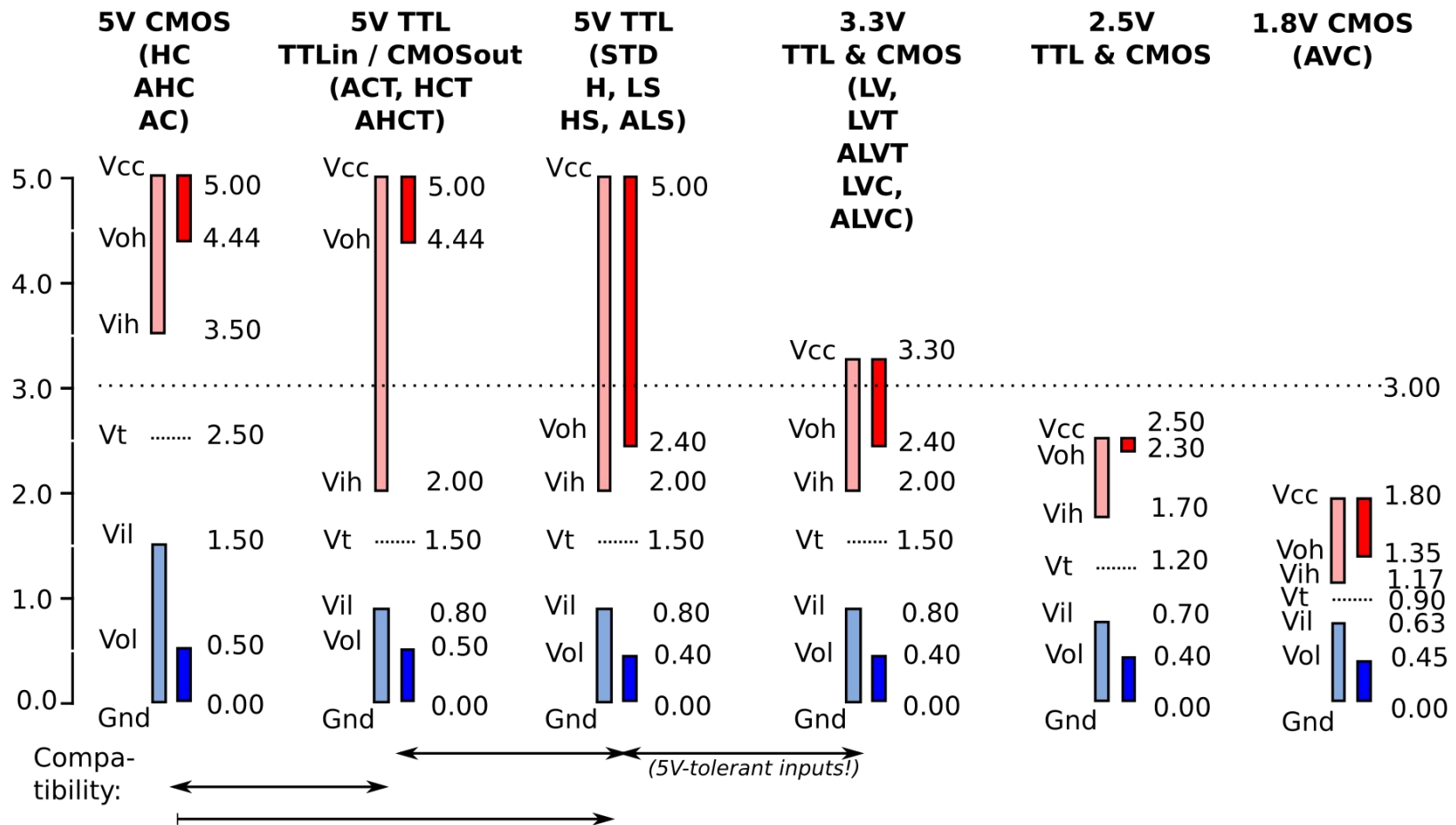
Voltage Levels

Ideal HIGH & LOW



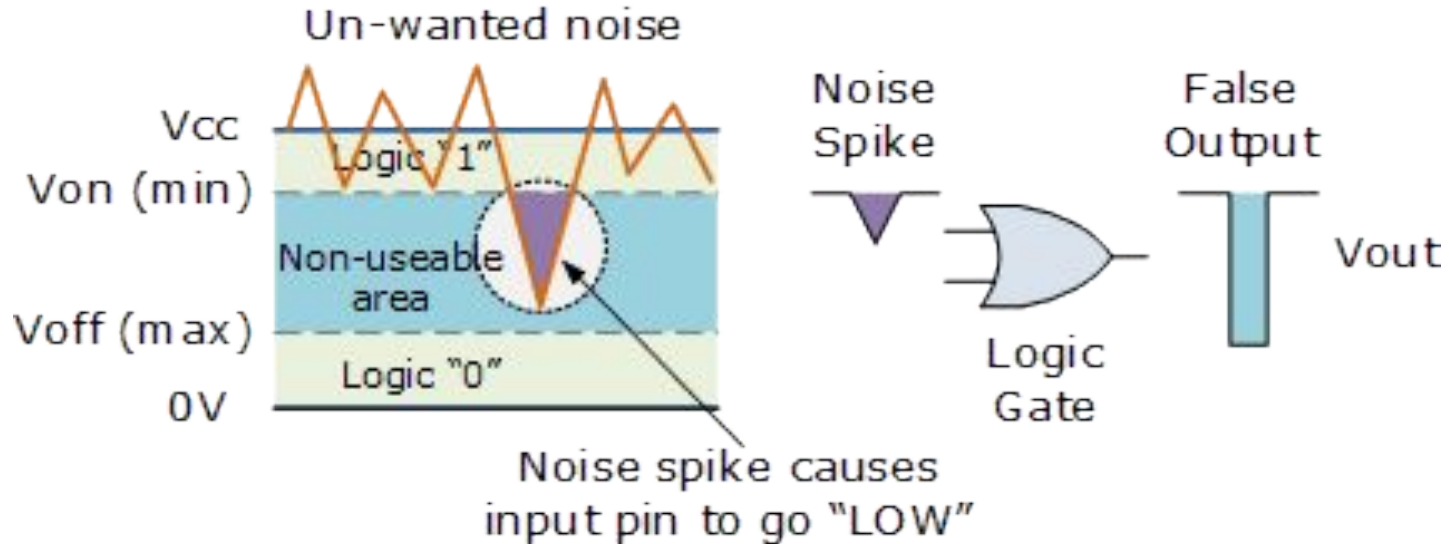
Device Type	Logic 0	Logic 1
TTL	0 to 0.8v	2.0 to 5v (V_{CC})
CMOS	0 to 1.5v	3.0 to 18v (V_{DD})

Voltage Levels



Data source: EETimes, A brief recap of popular logic standards (Mark Pearson, Maxim).

The Intermediate Region (“no-man’s land”)

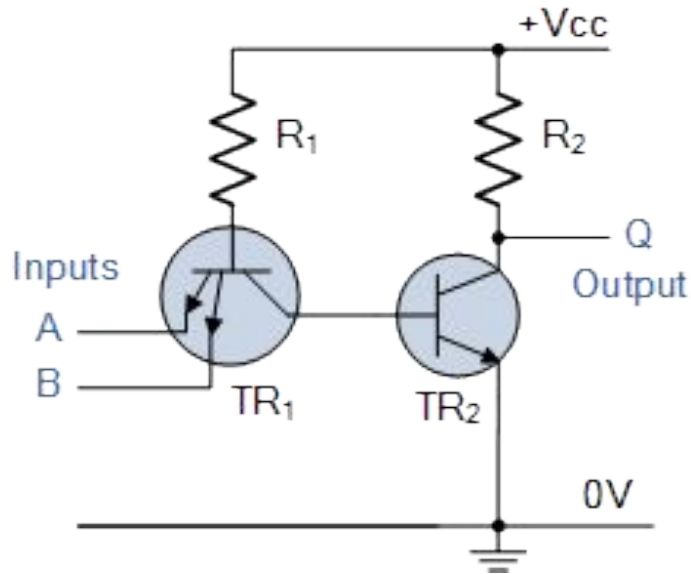


- Noise immunity of a gate
 - For a logic gate not to be affected by noise it must be able to tolerate a certain amount of unwanted noise on its input without changing the state of its output.

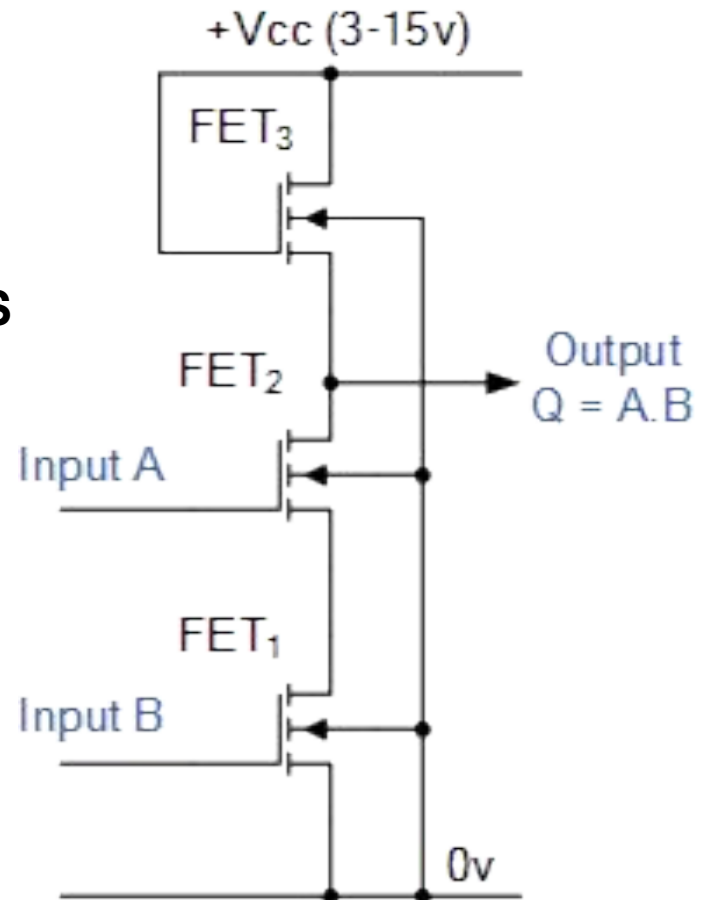
TTL/CMOS Logic Gates

2-input NAND Gate

TTL



CMOS

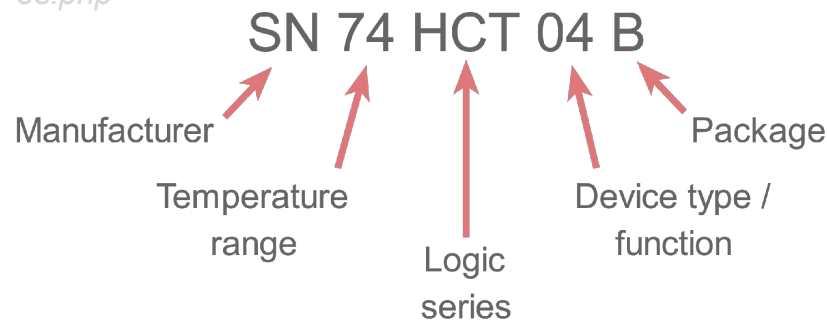


TTL/CMOS Logic Gates

- TTL Limitations:
 - Since bipolar transistors are current operated, their current consumption is very high
 - Limited operating speed due to slow rise/fall time (propagation delay)
- CMOS Strengths:
 - Near zero power consumption (1-2 μA)
 - High-speed switching ($>100\text{MHz}$)

TTL Numbering

https://www.electronics-notes.com/articles/electronic_components/logic-ic-families-technologies/ic-numbering-schemes.php

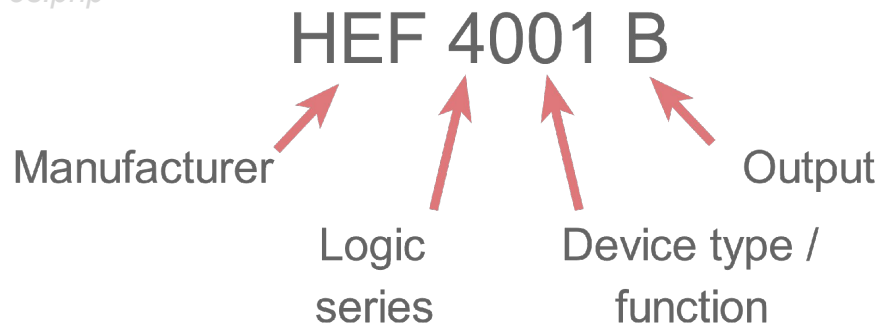


PART NUMBERING SCHEME FOR 74XX00 SERIES LOGIC ICs

Manufacturer	This code normally consists of two letters and is a code normally used by a given manufacturer. SN is one used by Texas Instruments. Other manufacturers have their own codes that they place here.
Temperature range	This is indicated by these two figures. 74 indicates 0°C to 70°C commercial and 54 military: -55°C to +125°C. For most applications the 74 series is perfectly acceptable and this series will be found in consumer devices.
Logic series	This is the sub-family. 7400 for example is the basic series, but there are many others.
Device	This indicates the device function / type. For example devices with 04 are hex inverters, etc. They are the common across all sub-families.
Package code	This is the package suffix. It is necessary to refer to the manufacturers datasheets as these codes vary between manufacturers.

CMOS Numbering

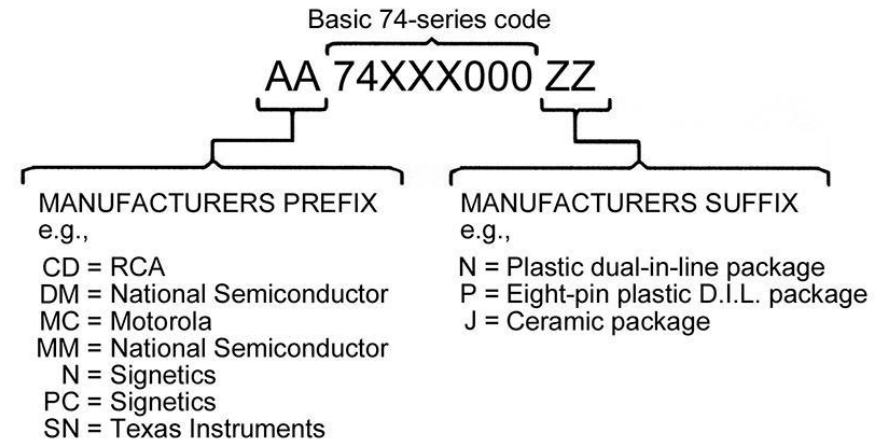
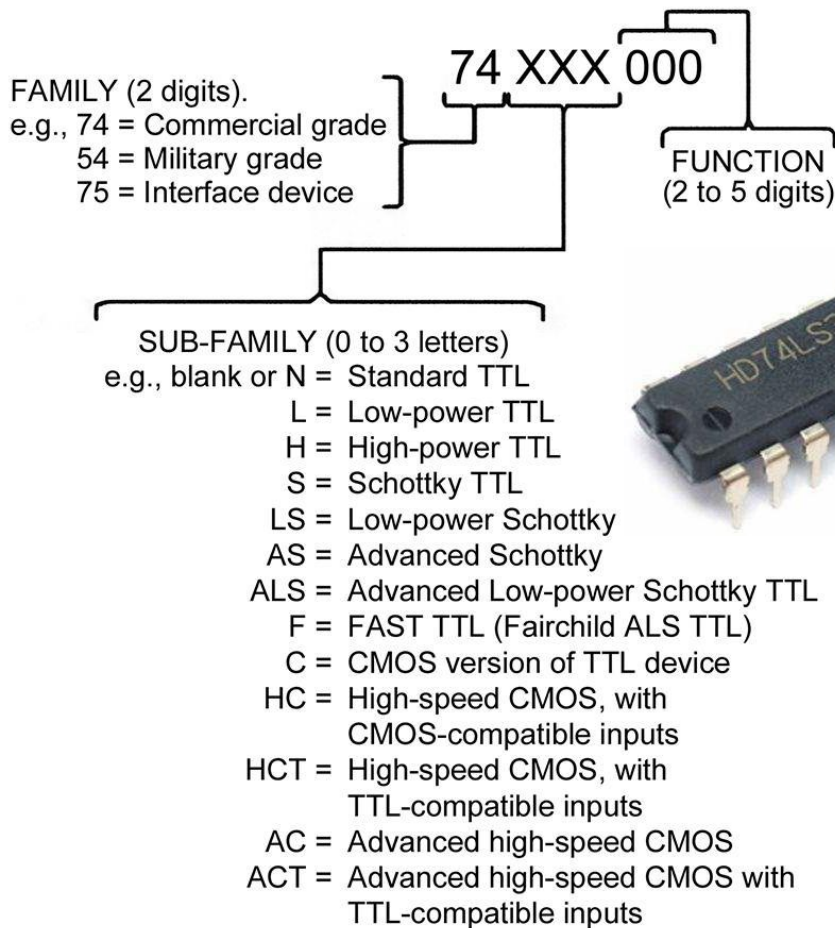
https://www.electronics-notes.com/articles/electronic_components/logic-ic-families-technologies/ic-numbering-schemes.php



PART NUMBERING SCHEME FOR 4000 SERIES LOGIC ICs

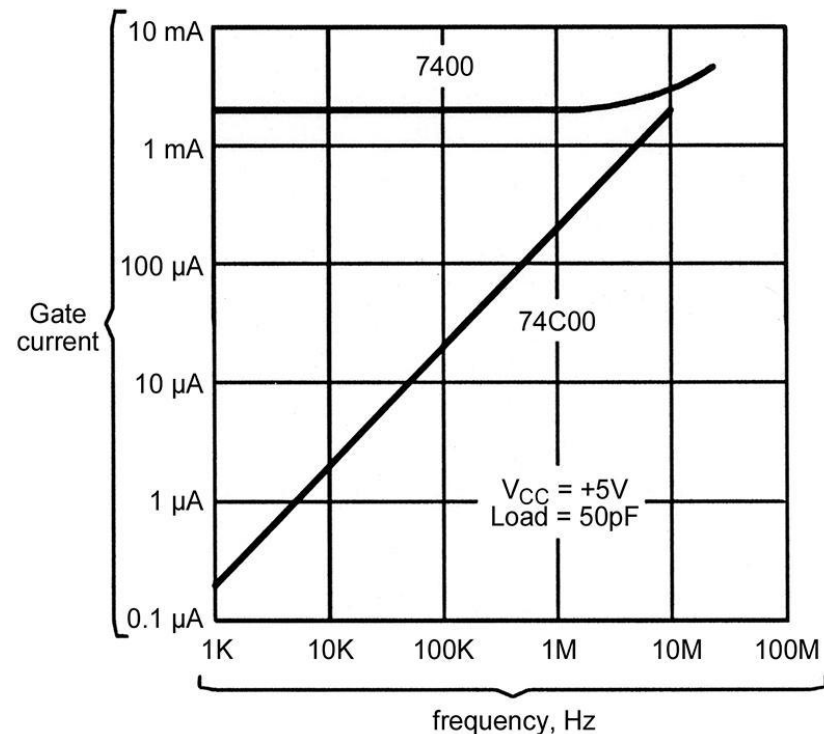
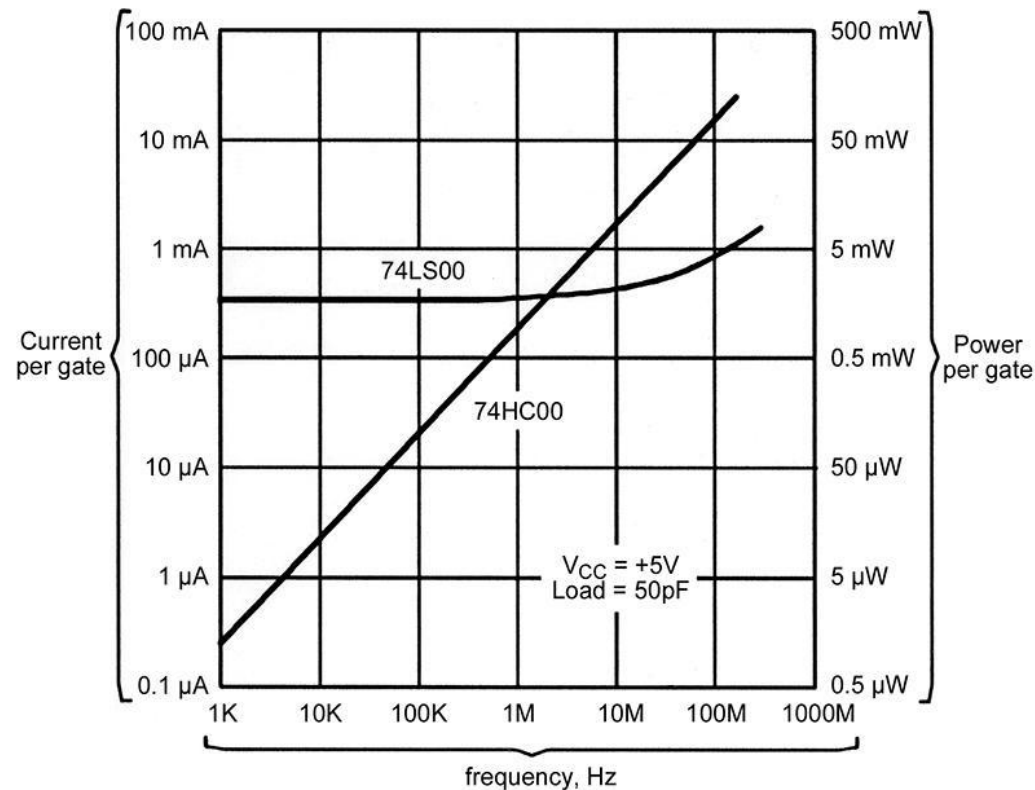
Manufacturer	This code normally consists of two letters and is a code normally used by a given manufacturer. HEF is what used to be Mullard / Phillips.
Series	This is indicated by these the 4 in the 4000 figure.
Device	This indicates the device function / type. For example devices the 01, i.e. 4001 is a quad 2-input NOR.
Output	Early devices did not have any output buffering and were lacking in drive. Later versions were buffered and these chips had the "B" suffix added to the part number.

TTL Sub-Families



PARAMETERS	74-Series TTL Sub-families							UNITS
	Standard	L	H	S	LS	AS	ALS	
Propagation Delay (2-input NAND gate)	9nS	33	6	3	8	2	4	nS
Power Dissipation (per gate)	10 mW	1	22	20	2	22	1	mW
V_{IH}	2.0V	2.0	2.0	2.0	2.0	2.0	2.0	V
V_{OH}	2.4V	2.4	2.4	2.7	2.7	$V_{DD}-2V$	$V_{DD}-2V$	V
NM-H	400 mV	400	400	700	700	700	700	mV
V_{IL}	0.8V	0.7	0.8	0.8	0.8	0.8	0.8	V
V_{OL}	0.4V	0.3	0.4	0.5	0.5	0.5	0.5	V
NM-L	400 mV	300	400	300	300	300	300	mV

Comparing the Frequency vs. Power Consumption



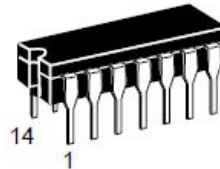
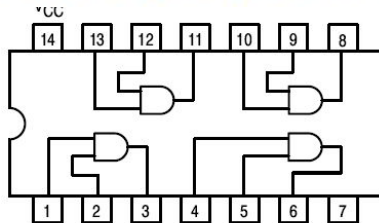
Inspecting Datasheets



MOTOROLA

QUAD 2-INPUT AND GATE

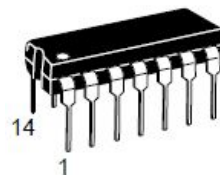
SN54/74LS08



J SUFFIX
CERAMIC
CASE 632-08



D SUFFIX
SOIC
CASE 751A-02



N SUFFIX
PLASTIC
CASE 646-06

ORDERING INFORMATION

SN54LSXXJ	Ceramic
SN74LSXXN	Plastic
SN74LSXXD	SOIC

GUARANTEED OPERATING RANGES

Symbol	Parameter		Min	Typ	Max	Unit
V _{CC}	Supply Voltage	54 74	4.5 4.75	5.0 5.0	5.5 5.25	V
T _A	Operating Ambient Temperature Range	54 74	-55 0	25 25	125 70	°C
I _{OH}	Output Current — High	54, 74			-0.4	mA
I _{OL}	Output Current — Low	54 74			4.0 8.0	mA

DC CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (unless otherwise specified)

Symbol	Parameter		Limits			Unit	Test Conditions	
			Min	Typ	Max			
V _{IH}	Input HIGH Voltage		2.0			V	Guaranteed Input HIGH Voltage for All Inputs	
V _{IL}	Input LOW Voltage	54			0.7	V	Guaranteed Input LOW Voltage for All Inputs	
		74			0.8			
V _{IK}	Input Clamp Diode Voltage			−0.65	−1.5	V	V _{CC} = MIN, I _{IN} = −18 mA	
V _{OH}	Output HIGH Voltage	54	2.5	3.5		V	V _{CC} = MIN, I _{OH} = MAX, V _{IN} = V _{IH} or V _{IL} per Truth Table	
		74	2.7	3.5		V		
V _{OL}	Output LOW Voltage	54, 74		0.25	0.4	V	I _{OL} = 4.0 mA	V _{CC} = V _{CC} MIN, V _{IN} = V _{IL} or V _{IH} per Truth Table
		74		0.35	0.5	V	I _{OL} = 8.0 mA	
I _{IH}	Input HIGH Current				20	μA	V _{CC} = MAX, V _{IN} = 2.7 V	
					0.1	mA	V _{CC} = MAX, V _{IN} = 7.0 V	
I _{IL}	Input LOW Current				−0.4	mA	V _{CC} = MAX, V _{IN} = 0.4 V	
I _{OS}	Short Circuit Current (Note 1)		−20		−100	mA	V _{CC} = MAX	
I _{CC}	Power Supply Current Total, Output HIGH				4.8	mA	V _{CC} = MAX	
	Total, Output LOW				8.8			

Note 1: Not more than one output should be shorted at a time, nor for more than 1 second.

AC CHARACTERISTICS ($T_A = 25^\circ\text{C}$)

Symbol	Parameter		Limits			Unit	Test Conditions	
			Min	Typ	Max			
t_{PLH}	Turn-Off Delay, Input to Output			8.0	15	ns	$V_{CC} = 5.0 \text{ V}$ $C_L = 15 \text{ pF}$	
t_{PHL}	Turn-On Delay, Input to Output			10	20	ns		

Inspecting Datasheets



August 1986
Revised July 2001

DM7408 Quad 2-Input AND Gates

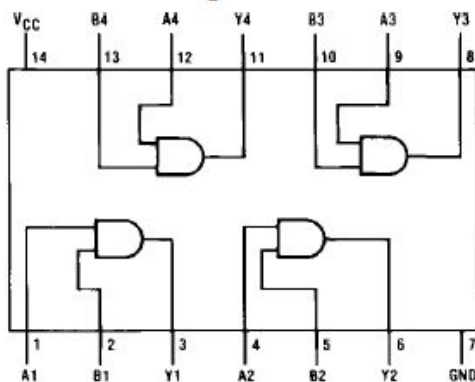
General Description

This device contains four independent gates each of which performs the logic AND function.

Ordering Code:

Order Number	Package Number	Package Description
DM7408N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide

Connection Diagram



Function Table

$$Y = AB$$

Inputs		Output
A	B	Y
L	L	L
L	H	L
H	L	L
H	H	H

H = HIGH Logic Level
L = LOW Logic Level

Recommended Operating Conditions

Symbol	Parameter	Min	Nom	Max	Units
V_{CC}	Supply Voltage	4.75	5	5.25	V
V_{IH}	HIGH Level Input Voltage	2			V
V_{IL}	LOW Level Input Voltage			0.8	V
I_{OH}	HIGH Level Output Current			-0.8	mA
I_{OL}	LOW Level Output Current			16	mA
T_A	Free Air Operating Temperature	0		70	°C

Electrical Characteristics

over recommended operating free air temperature range (unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ (Note 2)	Max	Units
V_I	Input Clamp Voltage	$V_{CC} = \text{Min}$, $I_I = -12 \text{ mA}$			-1.5	V
V_{OH}	HIGH Level Output Voltage	$V_{CC} = \text{Min}$, $I_{OH} = \text{Max}$ $V_{IL} = \text{Max}$	2.4	3.4		V
V_{OL}	LOW Level Output Voltage	$V_{CC} = \text{Min}$, $I_{OL} = \text{Max}$ $V_{IH} = \text{Min}$		0.2	0.4	V
I_I	Input Current @ Max Input Voltage	$V_{CC} = \text{Max}$, $V_I = 5.5 \text{ V}$			1	mA
I_{IH}	HIGH Level Input Current	$V_{CC} = \text{Max}$, $V_I = 2.4 \text{ V}$			40	μA
I_{IL}	LOW Level Input Current	$V_{CC} = \text{Max}$, $V_I = 0.4 \text{ V}$			-1.6	mA
I_{OS}	Short Circuit Output Current	$V_{CC} = \text{Max}$ (Note 3)	-18		-55	mA
I_{CCH}	Supply Current with Outputs HIGH	$V_{CC} = \text{Max}$		11	21	mA
I_{CCL}	Supply Current with Outputs LOW	$V_{CC} = \text{Max}$		20	33	mA

Note 2: All typicals are at $V_{CC} = 5 \text{ V}$, $T_A = 25^\circ\text{C}$.

Note 3: Not more than one output should be shorted at a time.

Switching Characteristics

at $V_{CC} = 5 \text{ V}$ and $T_A = 25^\circ\text{C}$

Symbol	Parameter	Conditions	Min	Max	Units
t_{PLH}	Propagation Delay Time LOW-to-HIGH Level Output	$C_L = 15 \text{ pF}$ $R_L = 400\Omega$		27	ns
t_{PHL}	Propagation Delay Time HIGH-to-LOW Level Output			19	ns